SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Case Study (20 Questions)

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1 – Imitation (BL4)	**Assessment:**	Assessment Tool 3 – Case Study
Weightage:	30 % of CIA	**Total Marks:**	20 (Answer ANY ONE) 1 x 20 = 20
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems. (BTL 4)*		
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Section / Batch:	Slot-A/2024-28	**Date:**	30-10-2026

Code of Conduct

I Ch.V.Ramayogi/192424319(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ch.V.Ramayogi

Instructions

- This test contains 20 questions (ANSWER ANY ONE), Total: 20 Marks.
- No calculators or electronic devices permitted.
- Duration: 30 minutes.

Questions

1. Network Congestion in a Festival Area

During a large festival, thousands of users gather in a small area. Mobile users experience call drops and slow data speeds. The network operator analyzes the issue and identifies high traffic load. Engineers implement cell splitting and sectoring to improve coverage. Additional temporary base stations are deployed. Frequency reuse planning is adjusted. Performance metrics such as call blocking probability are monitored. After optimization, service quality improves. This case highlights cellular network capacity planning. It also demonstrates real-time performance management.

ANSWER:

Problem Analysis:

During a large festival, thousands of mobile users are concentrated in a small geographical area. Many users try to make calls and access mobile data at the same time. This creates heavy traffic, call drops, slow data speeds and a higher probability of call blocking. The main problem is that the available network capacity is not sufficient for the sudden increase in demand.

Relevant Mobile Computing Concepts:

Cell splitting divides a large cell into smaller cells so that traffic can be handled by more base stations. Sectoring divides a cell into sectors using directional antennas, helping reduce interference and improve capacity. Frequency reuse allows the same frequency channels to be used in sufficiently separated cells. Temporary base stations provide additional capacity during the event.

Solution:

1. Study the expected number of users and traffic demand before the festival.
2. Deploy temporary base stations or mobile cells in crowded locations.
3. Split overloaded cells into smaller cells where practical.
4. Use sectoring and suitable frequency reuse planning.
5. Balance users and traffic among available cells.
6. Monitor call blocking probability, dropped-call rate and data throughput continuously.

Conclusion:

The combination of temporary base stations, cell splitting, sectoring and frequency planning increases network capacity. Continuous monitoring helps engineers identify overloaded areas and maintain reliable service during the festival.

2. Seamless Handoff in High-Speed Train

Passengers on a high-speed train experience frequent network switching. Calls drop as users move across multiple cells rapidly. Engineers implement soft handoff techniques. Base stations coordinate to maintain signal continuity. Mobile stations continuously measure signal strength. The network decides the best base station dynamically. Protocol stack signaling ensures smooth transitions. Call continuity improves significantly. The solution enhances user experience. This scenario highlights mobility management.

ANSWER:

Problem Analysis:

Passengers on a high-speed train move rapidly through the coverage areas of several cells. Their mobile devices must change from one serving base station to another. If the network does not complete this handoff quickly, calls may be interrupted and data sessions may experience delays.

Relevant Mobile Computing Concepts:

Handoff, also called handover, transfers an active connection from one cell or base station to another. The mobile station measures the signal strength of nearby cells. The network uses these measurements to select a suitable target base station. Signaling between the mobile station and network coordinates the

transition. Soft handoff, where supported by the cellular technology, can maintain a connection with more than one base station during a transition.

Solution:

1. Continuously measure the serving-cell and neighbouring-cell signal levels.
2. Configure suitable handoff thresholds to avoid late handoffs.
3. Coordinate neighbouring base stations so that handoff information is available quickly.
4. Reduce signaling and processing delays during the transition.
5. Provide adequate coverage along the railway route, including tunnels and stations.
6. Test call continuity and handoff failure rates at train speeds.

Conclusion:

Fast signal measurement, coordinated base stations and efficient handoff procedures reduce call drops. These measures allow passengers to maintain calls and data sessions while travelling at high speed.

3. Rural Network Coverage Deployment

A telecom operator plans to deploy a network in a rural area. The region has sparse population and wide coverage needs. Engineers design large cell sizes to reduce infrastructure cost. High-power base stations are installed. Frequency reuse is optimized to avoid interference. Mobile stations connect to distant towers. Performance is measured through coverage and signal strength. The network achieves cost-effective deployment. Users receive stable connectivity. This case highlights coverage planning.

ANSWER:

The rural area has a small population spread over a wide region. The operator must provide broad coverage without installing an unnecessarily large number of sites.

Design factor	Recommended consideration
Cell size	Use larger cells where terrain and radio conditions permit.
Base-station location	Choose elevated, suitable sites to cover villages and roads.
Transmission power	Use appropriately configured high-power base stations and antennas.

Frequency plan	Reuse frequencies carefully to avoid interference between distant cells.
Evaluation	Measure coverage, signal strength and service quality in the field.

Deployment sequence

Area survey → Site selection → Base-station installation → Frequency planning → Field testing

Conclusion: A carefully selected tower location and suitable large-cell design can provide cost-effective rural connectivity. Actual coverage must be verified through field measurements.

4. Indoor Coverage Problem in Shopping Mall

Users inside a shopping mall experience poor signal strength. Thick walls reduce signal penetration. Engineers deploy microcells inside the building. Small base stations improve signal quality. Mobile stations connect to nearest indoor cells. Handoff mechanisms maintain call continuity. Data speeds improve significantly. Performance metrics show reduced call drops. Customer satisfaction increases. This scenario demonstrates microcell deployment.

ANSWER:

Problem observed	**Technical reason**	**Solution**
Weak signal inside shops	Walls and building materials weaken radio signals.	Install indoor small cells or microcells.
Poor basement coverage	Outdoor signals may not penetrate underground levels.	Place indoor coverage equipment in the basement.
Call interruption while moving	Phone changes between indoor and outdoor coverage areas.	Configure suitable handoff between cells.
Slow indoor data	Weak signal or insufficient indoor capacity.	Plan indoor cells according to user density.

Implementation: First conduct an indoor signal survey. Then position small base stations in weak areas, configure power and frequency to limit interference, and test coverage in shops, corridors and common areas.

Conclusion: Microcells improve signal quality inside the mall and provide more reliable indoor service.

5. Call Drop During Movement

A user traveling in a car experiences call drops. The network struggles with fast mobility. Engineers analyze handoff delays. Base station coordination is improved. Mobile station signal monitoring is enhanced. Network protocol stacks handle faster transitions. Handoff time is reduced. Call continuity improves. User experience becomes smoother. This case highlights handoff optimization.

ANSWER:

User moves in a car → Serving-cell signal weakens → Handoff is delayed → Connection may fail → Call drops

Why the call drops

The mobile station is moving between cells. If the network does not identify and prepare the next suitable cell in time, the serving connection may be lost before the handoff is completed.

Corrective measures

Improve measurement of serving-cell and neighbouring-cell signal strength.

Tune handoff thresholds and timers to suit the mobility conditions.

Improve coordination between neighbouring base stations.

Reduce avoidable signaling and handoff processing delay.

Check for coverage gaps along the road.

Expected result

A faster and more reliable handoff reduces dropped calls and gives users a smoother experience while travelling.

6. Network Overload in Business District

A business district experiences heavy traffic during working hours. Many users access mobile data simultaneously. Network capacity becomes insufficient. Engineers implement sectoring. Additional base stations are deployed. Frequency reuse increases system capacity. Mobile stations distribute across sectors. Performance improves significantly. Network congestion reduces. This case demonstrates capacity improvement.

ANSWER:

The business district has heavy simultaneous voice and data usage during working hours. The existing cell capacity is insufficient, so the operator should increase capacity and distribute traffic more effectively.

Proposed measures

Identify the busiest cells and peak usage periods from traffic measurements.

Deploy additional base stations in locations with persistent overload.

Apply sectoring to divide a cell's coverage into directional sectors.

Optimize frequency reuse so that capacity increases without unacceptable interference.

Use load balancing to distribute users among available cells or sectors.

How success will be measured

Metric	**Desired observation**
Cell utilization	Less persistent overload during busy hours.
Call blocking	Fewer call attempts rejected due to unavailable resources.
Dropped-call rate	Fewer calls interrupted unexpectedly.
Data throughput	Improved user data speeds.

Conclusion: More radio capacity, sectoring and load balancing address the business district's peak-hour congestion.

7. International Roaming Scenario

A user travels to another country. The mobile station connects to a foreign network. The Visitor Location Register updates user location. Home Location Register verifies subscriber details. Authentication protocols ensure security. Calls and data services remain available. Billing is handled through roaming agreements. Performance remains stable. User experiences seamless connectivity. This case highlights roaming architecture.

ANSWER:

When a subscriber travels abroad, the phone may register with a visited mobile network. The home network and visited network exchange information so that the subscriber can be authenticated and provided with permitted services.

Component	Function
Mobile station	Searches for and registers with a compatible visited network.
Visited network	Provides radio access and serving network functions in the foreign country.
VLR	Keeps temporary information about subscribers currently visiting its service area.
HLR	Maintains permanent subscriber information in the home network.
Authentication	Checks the subscriber's identity before access is granted.
Roaming agreement	Defines how participating operators provide service and handle usage records.

Phone selects visited network → Registration request → Subscriber information checked → Authentication → Service access

Conclusion: HLR, VLR, authentication procedures and roaming agreements enable subscribers to use mobile services outside their home country.

8. Emergency Communication Overload

During a natural disaster, users make emergency calls. Network becomes congested. Engineers prioritize emergency services. Traffic management protocols are activated. Additional mobile base stations are deployed. Network capacity increases temporarily. Mobile stations connect to available cells. Call success rate improves. Emergency communication is maintained. This scenario highlights emergency network planning.

ANSWER:

During a disaster, many people attempt to call at the same time. Network congestion can prevent important calls from connecting. The response must protect emergency communication and restore capacity.

Priority actions

Identify the affected network sites and the areas where emergency communication is essential.

Activate supported priority or preferential access arrangements for authorized emergency services.

Deploy temporary mobile base stations where existing infrastructure is damaged or overloaded.

Restore power and backhaul links to essential sites.

Coordinate with emergency agencies to identify communication needs.

Monitor network availability and call success during the response.

Expected benefits

Improved access for emergency responders.

Additional temporary coverage and capacity.

Faster identification of network failures.

Better continuity of essential communication.

Conclusion: Emergency priority arrangements and temporary infrastructure help maintain essential communications during disasters.

9. High Interference Between Cells

Users experience poor signal quality. Engineers detect co-channel interference. Frequency planning is optimized. Cell sectoring is introduced. Power control is implemented. Mobile stations switch to better signals. Interference reduces significantly. Call quality improves. Network efficiency increases. This case demonstrates interference management.

ANSWER:

The reported poor signal quality may be caused by co-channel interference. This occurs when cells using the same frequency are close enough for their signals to interfere.

Diagnosis and correction

Stage	Action
1. Detect	Measure signal quality and identify affected cells and frequencies.
2. Analyze	Check whether nearby cells using the same channel are causing interference.
3. Plan	Review frequency reuse and change unsuitable channel assignments.
4. Optimize	Adjust sector antennas, antenna direction and transmission power.
5. Verify	Measure signal quality, call quality and data performance again.

Conclusion: Better frequency planning, sectoring and power control reduce co-channel interference and improve network efficiency.

10. Mobile Registration Failure

A user switches on a phone but cannot connect. Mobile station sends registration request. Network authentication fails. Engineers check HLR and VLR databases. SIM verification is performed. Protocol stack signaling is analyzed. Issue is resolved. Mobile station connects successfully. Network service resumes. This case highlights registration procedures.

ANWER:

A phone that cannot register may have a SIM issue, an authentication failure, incorrect subscriber records or a signaling problem.

Troubleshooting checklist

SIM: Confirm that the SIM is correctly inserted and active.

Network selection: Check whether the phone detects a compatible network.

Authentication: Inspect whether subscriber verification succeeds.

HLR/VLR: Verify that subscriber and temporary location records are correct.

Account status: Check for service restrictions or barring.

Signaling: Review registration messages to find where the procedure fails.

Registration flow

SIM detected → Network selected → Registration request → Authentication → Registration accepted

Conclusion: Checking the SIM, subscriber databases and signaling procedure helps locate the registration failure and restore network access.

11. Airport Network Congestion

Airport terminals have dense user traffic. Many users stream video content. Network experiences congestion. Engineers deploy small cells. Load balancing is introduced. Protocol optimization is applied. Mobile stations distribute across cells. Performance improves significantly. Data speeds increase. This case demonstrates high-density network design.

ANSWER:

Airport terminals have dense user traffic because passengers stream videos, make calls and use online services while waiting. Congestion is likely to be concentrated in departure halls, boarding gates and waiting areas.

Deployment plan

Map the busiest terminal areas and measure existing network load.

Install small cells in high-demand indoor locations.

Use load balancing to distribute mobile stations among available cells.

Optimize radio and signaling parameters.

Coordinate frequency assignments to reduce interference between nearby cells.

Monitor throughput, congestion and call quality during busy travel periods.

Conclusion: Small cells increase local capacity, while load balancing helps prevent individual airport cells from becoming overloaded.

12. Mobile Data Delay Issue

Users complain about slow internet speeds. Engineers analyze protocol stack delays. Network signaling optimization is performed. Base station configuration is adjusted. Mobile stations reconnect efficiently. Data transmission improves. Performance metrics show improvement. Customer satisfaction increases. This case highlights protocol optimization.

ANSWER:

Protocol-layer performance explanation

Slow mobile internet may result from weak radio conditions, congestion, signaling delays or errors in the protocol stack. Engineers should measure the delay and investigate the communication path.

Area to inspect	What engineers check
Radio link	Signal quality and retransmission-related problems.
Link and network layers	Transmission errors and packet delivery problems.
Signaling	Delay in establishing or managing a connection.
Base-station settings	Radio configuration and available capacity.
Performance	Latency, throughput and packet loss.

Verification

After correcting the identified issue, repeat the same data-transfer tests under comparable conditions. Compare latency, throughput and packet loss with the original measurements.

Conclusion: Layer-by-layer analysis and radio optimization help identify the source of mobile data delay.

13. University Campus Coverage

Students experience weak signals across campus. Engineers deploy multiple small cells. Handoff between cells is optimized. Mobile stations switch seamlessly. Network capacity improves. Data performance increases. Call drop reduces. Coverage becomes uniform. This case demonstrates campus network planning.

ANSWER:

Campus coverage plan

A campus includes classrooms, laboratories, libraries, hostels and open spaces. One cell may not provide uniform coverage in all these locations, so multiple small cells may be needed.

Suggested coverage zones

Campus location	Network planning focus
Classrooms and laboratories	Indoor signal strength and capacity during class hours.
Library	Reliable indoor coverage for study and online access.
Hostels	Capacity for high evening data usage.
Open areas	Continuous coverage between buildings.
Campus roads and walkways	Smooth handoff while users move between cells.

The operator should survey weak-signal locations, deploy small cells where required, coordinate frequencies and optimize handoff between neighbouring cells.

Conclusion: Multiple appropriately planned cells provide more uniform coverage and reduce call drops across the university campus.

14. Stadium Event Traffic Management

A stadium hosts a major event. Thousands of users access mobile networks. Temporary base stations are installed. Frequency reuse is optimized. Mobile stations connect dynamically. Network traffic is balanced. Performance remains stable. Call drops reduce. This case highlights event-based deployment.

ANSWER:

Before / during / after event plan

Time	Network activity
Before event	Estimate audience size, forecast traffic and identify high-density seating areas.
Before event	Install temporary base stations and plan sectors and frequency reuse.
During event	Monitor traffic, balance users and adjust network settings where needed.

During event	Check call drops, data throughput and cell utilization.
After event	Review performance measurements and improve the plan for future events.

A stadium creates a concentrated and temporary demand peak. Normal nearby cells may not be designed for thousands of spectators using data at the same time.

Conclusion: Temporary base stations, careful frequency planning and real-time traffic balancing help maintain stable service during large events.

15. Mobile Station Battery Optimization

A mobile user experiences fast battery drain. Engineers analyze signal searching frequency. Network adjusts power control. Mobile station reduces transmission power. Protocol stack optimization is applied. Battery life improves. Network efficiency increases. This case highlights mobile station optimization.

ANSWER:

Factors and improvements

Battery-drain factor	**Possible improvement**
Repeated network searching	Improve coverage in areas where the device repeatedly loses service.
Weak signal	Improve radio coverage and avoid unnecessary high-power transmission.
High transmit power	Use suitable network power-control settings.
Frequent signaling	Optimize procedures to reduce unnecessary signaling activity.
Unstable connection	Investigate coverage and connection failures that cause repeated reconnection.

The mobile station is the user's phone and radio interface. Battery usage should be compared under similar signal and usage conditions before and after network changes.

Conclusion: Better coverage, efficient power control and reduced unnecessary signaling can help improve mobile-station battery life.

16. Urban High-Rise Coverage

Users in tall buildings face signal issues. Engineers deploy distributed antenna systems. Microcells are installed. Mobile stations connect to nearest antenna. Signal strength improves. Data speeds increase. Coverage becomes consistent. This case highlights vertical coverage planning.

ANSWER:

Compare two indoor coverage approaches

Feature	**Distributed Antenna System (DAS)**	**Microcells**
Purpose	Distributes a signal through multiple antennas across a building.	Provides localized cellular coverage and capacity.
Typical use	Large buildings requiring coverage across many floors or areas.	Specific locations with weak coverage or high user demand.
Planning need	Antenna placement and distribution throughout the building.	Selection of suitable locations and cell configuration.
Important concern	Coordination of indoor and outdoor signals.	Interference and handoff with neighbouring cells.

Recommended approach

Conduct a floor-by-floor survey. Use a distributed antenna system when broad building-wide distribution is required, and consider microcells for localized coverage or capacity needs. Coordinate indoor and outdoor radio settings.

Conclusion: DAS and microcells can improve high-rise indoor coverage when selected and configured according to the building's needs.

17. Network Expansion Planning

A telecom company expands coverage. Engineers analyze traffic demand. New base stations are deployed. Frequency reuse planning is updated. Mobile stations connect to new cells. Performance improves. Network capacity increases. This case highlights expansion planning.

ANSWER:

Expansion lifecycle

Traffic study → Coverage-gap analysis → Site selection → Base-station deployment → Frequency and handoff planning → Performance testing

Explanation of the stages

Traffic study: estimate present and future user demand.

Coverage analysis: locate areas with weak or missing service.

Site selection: choose base-station locations based on terrain and user distribution.

Deployment: install and configure the new base stations.

Integration: update frequency reuse and handoff arrangements with neighbouring cells.

Testing: measure coverage, capacity, call quality and data throughput.

Conclusion: A planned expansion increases coverage and capacity while reducing the risk of interference with existing cells.

18. Protocol Stack Failure

Users face connectivity issues. Engineers inspect protocol layers. Signaling errors are identified. Stack configuration is corrected. Mobile stations reconnect. Data services resume. Performance stabilizes. This case highlights protocol stack troubleshooting.

ANSWER:

Layer-by-layer troubleshooting

A protocol stack contains layers that work together to transfer data. A failure in one layer or an incorrect configuration can prevent the mobile station from connecting or using data services.

Investigation area	**Troubleshooting question**
Physical / radio communication	Is the mobile device receiving an adequate radio signal?

Link functions	Are link-level exchanges completing correctly?
Network delivery	Are network-layer messages and packet delivery working?
Signaling procedures	Are registration and connection-management messages successful?
Configuration	Are protocol parameters configured correctly?

Recovery sequence

Collect logs → Identify failing layer → Correct configuration → Reconnect device → Test services

Conclusion: Layer-by-layer diagnosis identifies the source of a protocol-stack failure and supports a targeted correction.

19. Smart City Network Deployment

A smart city requires reliable mobile connectivity. Engineers design dense cell deployment. Network architecture supports IoT devices. Mobile stations connect efficiently. Data traffic is optimized. Performance criteria are monitored. Coverage becomes reliable. This case highlights smart city deployment.

ANSWER:

Requirements and network architecture

Main requirements

Reliable coverage for residents and public services.

Connectivity for IoT devices used in city applications.

Capacity for ordinary mobile users and connected devices.

Efficient traffic handling across dense urban areas.

Authentication and security for connected devices.

Continuous monitoring of network performance.

Simplified architecture

IoT devices and mobile users → Cellular base stations / small cells → Mobile network infrastructure → Smart-city applications

Deployment approach: Identify device locations and traffic needs, install suitable base stations and small cells, plan frequency reuse, provide appropriate security procedures and monitor availability.

Conclusion: A well-planned cellular network supports smart-city services by connecting mobile users and IoT devices with reliable coverage and capacity.

20. Call Blocking Reduction

Users face call blocking during peak hours. Engineers analyze traffic patterns. Additional channels are allocated. Sectoring improves capacity. Mobile stations connect successfully. Call blocking probability reduces. Performance improves. This case highlights traffic management.

ANSWER:

Short analytical note with key metric

Call blocking happens when a new call cannot be established because the required network resources are unavailable. During peak hours, the number of call attempts may exceed the capacity of the cell.

Key term

Call blocking probability is the likelihood that a call attempt is rejected because the network cannot allocate the required resources.

Measures to reduce blocking

Measure call blocking during busy hours and identify overloaded cells.

Analyze traffic patterns and determine where extra capacity is needed.

Allocate available channels and radio resources efficiently.

Apply sectoring where appropriate to improve cell capacity.

Add base stations or additional capacity in locations with persistent overload.

Measure call blocking again to confirm that the changes are effective.

Result

When capacity and resource allocation improve, fewer call attempts are rejected. The operator should continue monitoring call blocking and dropped-call rates to maintain service quality.

Conclusion: Efficient channel allocation, sectoring and additional capacity reduce call blocking during peak usage.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____